3

Pressure Drop due to Friction

In this chapter, we introduce the concept of pressure in a liquid and how it is measured. The liquid flow velocity in a pipe, types of flow, and the importance of the Reynolds number will be discussed. For different flow regimes, such as laminar, critical, or turbulent, methods will be discussed as to how to calculate the pressure drop due to friction. Several popular formulas such as the Colebrook-White and Hazen-Williams equations will be presented and compared. Also we will cover minor pressure losses in piping such as those due to fittings and valves, and those resulting from pipe enlargements and contractions. We will also explore drag reduction as a means of reducing energy loss in pipe flow.

3.1 Pressure

Hydrostatics is the study of hydraulics that deals with liquid pressures and forces resulting from the weight of the liquid at rest. Although this book is mainly concerned with flow of liquids in pipelines, we will address some issues related to liquids at rest in order to discuss some fundamental issues pertaining to liquids at rest and in motion.

The force per unit area at a certain point within a liquid is called the pressure, p. This pressure at a certain depth, h, below the free surface of the liquid consists of equal pressures in all directions. This is known as Pascal's law. Consider an imaginary flat surface within the liquid located at a depth, h, below the liquid surface as shown in Figure 3.1. The pressure on this surface must act normal to the surface at all points along the surface because liquids at rest cannot transmit shear. The variation of pressure with the depth of the liquid is calculated by considering forces acting on a thin vertical cylinder of height Δh and a cross-sectional area Δa as shown in Figure 3.1.

Since the liquid is at rest, the cylindrical volume is in equilibrium due to the forces acting upon it. By the principles of statics, the algebraic sum of all forces acting on this cylinder in the vertical and horizontal directions must equal zero. The vertical forces on the cylinder consists of the weight of the cylinder and the forces due to liquid pressure P_1 at the top and P_2 at the bottom, as shown in Figure 3.1. Since the specific weight of the liquid, γ, does not change with pressure, we can write the following equation for the summation of forces in the vertical direction:

$$P_2\Delta a=\gamma\Delta h\Delta a+P_1\Delta a$$

where the term $\gamma\Delta h\Delta a$ represents the weight of the cylindrical element. Simplifying the above we get

$$P_2=\gamma\Delta h+P_1 \tag{3.1}$$

If we now imagine that the cylinder is extended to the liquid surface, P_1 becomes the pressure at the liquid surface (atmospheric pressure P_a) and Δh becomes h, the depth of the point in the liquid where the pressure is P_2. Replacing P_2 with P, the pressure in the liquid at depth h, Equation (3.1)

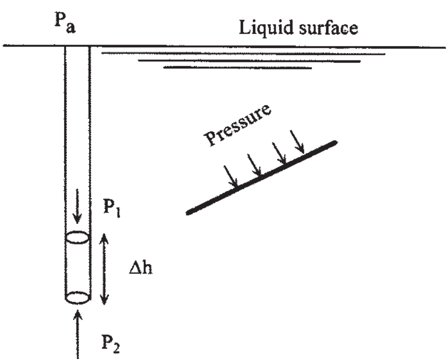

Figure 3.1 Pressure in a liquid.

becomes

$$P=\gamma h+P_a \tag{3.2}$$

From Equation (3.2) we conclude that the pressure in a liquid at a depth h increases with the depth. If the term P_a (atmospheric pressure) is neglected we can state that the gauge pressure (based on zero atmospheric pressure) at a depth h is simply γh. Therefore, the gauge pressure is

$$P=\gamma h \tag{3.3}$$

Dividing both sides by γ and transposing we can write

$$h=P/\gamma \tag{3.4}$$

In Equation (3.4) the term h represents the "pressure head" corresponding to the pressure P. It represents the depth in feet of liquid of specific weight γ required to produce the pressure P. Values of absolute pressure ($P+P_a$) are always positive whereas the gauge pressure P may be positive or negative depending on whether the pressure is greater or less than the atmospheric pressure. Negative gauge pressure means that a partial vacuum exists in the liquid.

From the above discussion it is clear that the absolute pressure within a liquid consists of the head pressure due to the depth of liquid and the atmospheric pressure at the liquid surface. The atmospheric pressure at a geographic location varies with the elevation above sea level. Because the density of the atmospheric air varies with the altitude, a straight-line relationship does not exist between the altitude and the atmospheric pressure (unlike the linear relationship between liquid pressure and depth). For most purposes, we can assume that the atmospheric pressure at sea level is approximately 14.7 psi in English units, or approximately 101 kPa in SI units.

The instrument used to measure the atmospheric pressure at a given location is called a barometer. A typical barometer is shown in Figure 3.2. In such an instrument the tube is filled with a heavy liquid (usually mercury) then quickly inverted and positioned in a container full of the liquid as shown in Figure 3.2. If the tube is sufficiently long, the level of liquid will fall slightly to cause a vapor space at the top of the tube just above the liquid surface. Equilibrium will be reached when the liquid vaporizes in the vapor space and creates a pressure P_v Because the density of mercury is high (approximately 13 times that of water) and its vapor pressure is low, it is an ideal liquid for a barometer. If a liquid such as water were used, a rather long tube would be needed to measure the atmospheric pressure, as we shall see shortly.

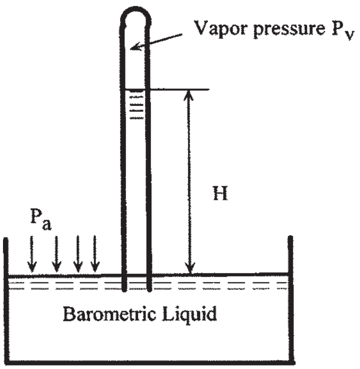

Figure 3.2 Barometer for measuring pressure.

From Figure 3.2 the atmospheric pressure P_a exerted at the surface of the liquid is equal to the sum of the vapor pressure P_v and the pressure generated by the column of the barometric liquid of height H_b.

$$P_a = P_v + \gamma H_b \tag{3.5}$$

where

P_a=Atmospheric pressure
P_v=Vapor pressure of barometric liquid
γ=Specific weight of barometric liquid
H_b=Barometric reading

In Equation (3.5), if pressures are in psi and liquid specific weight is in lb/ft^3, the pressures must be multiplied by 144 to obtain the barometric reading in feet of liquid.

Equation (3.5) is valid for barometers with any liquid. Since the vapor pressure of mercury is negligible we can rewrite Equation (3.5) for a mercury barometer as follows:

$$P_a = \gamma H_b \tag{3.6}$$

Let us compare the use of water and mercury as barometric liquids to measure the atmospheric pressure.

Example Problem 3.1

Assume the vapor pressure of water at 70°F is 0.3632 psi and its specific weight is 62.3 lb/ft^3. Mercury has a specific gravity of 13.54 and negligible vapor pressure. The sea level atmospheric pressure is 14.7 psi. Determine the barometric heights for water and mercury.

Solution

From Equation (3.5) for water,

$$14.7=0.3632+(62.3/144)\ H_b$$

The barometric height for water is

$$H_b=(14.7-0.3632)\times 144/62.3=33.14\ \text{ft}$$

Similarly, for mercury, neglecting the vapor pressure, using Equation (3.6) we get

$$14.7\times 144=(13.54\times 62.3)H_b$$

The barometric height for mercury is

$$H_b=(14.7\times 144)/(13.54\times 62.3)=2.51\ \text{ft}$$

It can be seen from the above that the mercury barometer requires a much shorter tube than a water barometer.

Manometers are instruments used to measure pressure in reservoirs, channels, and pipes. Manometers are discussed further in Chapter 10.

The pressure in a liquid is measured in lb/in.^{2} (psi) in English units or kilopascal (kPa) in SI units. Since pressure is measured using a gauge and is relative to the atmospheric pressure at the specific location, it is also reported as psig (psi gauge). The absolute pressure in a liquid is the sum of the gauge pressure and the atmospheric pressure at the location. Thus,

$$\text{Absolute pressure in psia}=\text{Gauge pressure in psig}+\text{Atmospheric pressure}$$

For example, if the pressure gauge reading is 800 psig, the absolute pressure in the liquid is

$$P_{abs}=800+14.7=814.7\ \text{psia}$$

This is based on the assumption that atmospheric pressure at the location is 14.7 psia.

Pressure in a liquid may also be referred to in terms of feet (or meters in SI units) of liquid head. By dividing the pressure in lb/ft by the liquid specific weight in lb/ft^3, we get the pressure head in feet of liquid. When expressed this way, the head represents the height of the liquid column required to match the given pressure in psig. For example, if the pressure in a liquid is 1000 psig, the head of liquid corresponding to this pressure is calculated as follows:

$$\text{Head}=2.31(\text{psig})/\text{Spgr ft (English units)} \tag{3.7}$$

$$\text{Head}=0.102(\text{kPa})/\text{Spgr m (SI units)} \tag{3.8}$$

where
Spgr=Liquid specific gravity
The factor 2.31 in Equation (3.7) comes from the ratio

$$\frac{144\ \text{in.}^2/\text{ft}^2}{62.34\ \text{lb}/\text{ft}^3}$$

where 62.34 lb/ft^3 is the specific weight of water. Therefore, if the liquid specific gravity is 0.85, the equivalent liquid head is

$$\text{Head}=(1000)(2.31)/0.85=2717.65\ \text{ft}$$

This means that the liquid pressure of 1000 psig is equivalent to the pressure exerted at the bottom of a liquid column, of specific gravity 0.85, that is 2717.65 feet high. If such a column of liquid had a cross-sectional area of 1 square inch, the weight of the column would be

$$2717.65(1/144)(62.34)(0.85)=1000\ \text{lb}$$

where 62.34 lb/ft^3 is the density of water. As the above weight acts on an area of 1 square inch, the calculated pressure is therefore 1000 psig.

We can analyze head pressure due to a column of liquid in another way: Consider a cylindrical column of liquid, of height H ft and cross-sectional area A in.^2. If the top surface of the liquid column is open to the atmosphere, we can calculate the pressure exerted by this column of liquid at its base as

$$\text{Pressure}=\frac{\text{Weight of liquid column}}{\text{Area of cross-section}}$$

or

$$\text{Pressure}=\frac{(\text{Volume}\times\text{Specific weight})}{\text{Area of cross-section}}$$
$$=(\text{AH}\gamma)/(144\times\text{A})$$

or

$$\text{Pressure}=\text{H}\gamma/144 \tag{3.9}$$

where
γ=Specific weight of liquid, lb/ft^3
The factor 144 is used to convert from in.^2 to ft^2.

If we use 62.34 lb/ft^3 for the specific weight of water, the pressure of a water column calculated from Equation (3.9) is

$$\text{Pressure} = H \times 62.34/144 = H/2.31$$

3.2 Velocity

Velocity of flow in a pipeline is the average velocity based on the pipe diameter and liquid flow rate. It may be calculated as follows:

$$\text{Velocity} = \text{Flow rate}/\text{Area of flow}$$

Depending on the type of flow (laminar, turbulent, etc.), the liquid velocity in a pipeline at a particular pipe cross-section will vary along the pipe radius. The liquid molecules at the pipe wall are at rest and therefore have zero velocity. As we approach the centerline of the pipe, the liquid molecules are increasingly free and therefore have increasing velocity. The variations in velocity for laminar flow and turbulent flow are as shown in Figure 3.3. In laminar flow (also known as viscous or streamline flow), the variation in velocity at a pipe cross-section is parabolic. In turbulent flow there is an approximate trapezoidal shape to the velocity profile.

If the units of flow rate are bbl/day and pipe inside diameter is in inches the following equation for average velocity may be used:

$$V = 0.0119(\text{bbl/day})/D^2 \tag{3.10}$$

where

V=Velocity, ft/s

D=Pipe internal diameter, in.

Other forms of the equation for velocity in different units are as follows:

$$V = 0.4085(\text{gal/min})/D^2 \tag{3.11}$$

$$V = 0.2859(\text{bbl/hr})/D^2 \tag{3.12}$$

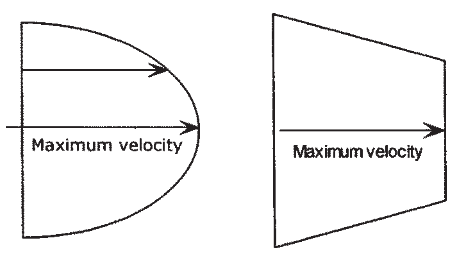

Figure 3.3 Velocity variation in a pipe for laminar flow (left) and turbulent flow (right).

where

V=Velocity, ft/s

D=Pipe internal diameter, in.

In SI units, the velocity is calculated as follows:

$$V=353.6777(m^3/hr)/D^2 \tag{3.13}$$

where

V=Velocity, m/s

D=Pipe internal diameter, mm

For example, liquid flowing through a 16 in. pipeline (wall thickness 0.250 in.) at the rate of 100,000 bbl/day has an average velocity of

$$0.0119(100{,}000)/(15.5)^2=4.95 \text{ ft/s}$$

This represents the average velocity at a particular cross-section of pipe. The velocity at the centerline will be higher than this, depending on whether the flow is turbulent or laminar.

3.3 Reynolds Number

Flow in a liquid pipeline may be smooth, laminar flow (also known as viscous or streamline flow). In this type of flow the liquid flows in layers or laminations without causing eddies or turbulence. If the pipe were transparent and we injected a dye into the flowing stream, it would flow smoothly in a straight line confirming smooth or laminar flow. As the liquid flow rate is increased, the velocity increases and the flow will change from laminar flow to turbulent flow with eddies and disturbances. This can be seen clearly when a dye is injected into the flowing stream.

An important dimensionless parameter called the Reynolds number is used in classifying the type of flow in pipelines. The Reynolds number of flow, R, is calculated as follows:

$$R=VD\rho/\mu \tag{3.14}$$

where

V=Average velocity, ft/s

D=Pipe internal diameter, ft

ρ=Liquid density, slugs/ft^3

μ=Absolute viscosity, lb-s/ft^2

R=Reynolds number, dimensionless

Since the kinematic viscosity $\nu=\mu/\rho$ the Reynolds number can also be expressed as

$$R=VD/\nu \tag{3.15}$$

where

ν=Kinematic viscosity, ft^2/s

Care should be taken to ensure that proper units are used in Equations (3.14) and (3.15) such that R is dimensionless.

Flow through pipes is classified into three main flow regimes:

1. Laminar flow: $R<2000$
2. Critical flow: $R>2000$ and $R<4000$
3. Turbulent flow: $R>4000$

Depending upon the Reynolds number, flow through pipes will fall in one of the above three flow regimes. Let us first examine the concepts of Reynolds number. Sometimes an R value of 2100 is used as the limit of laminar flow.

Using the customary units of the pipeline industry, the Reynolds number can be calculated using the following formula:

$$R=92.24\ Q/(\nu D) \tag{3.16}$$

where

Q=Flow rate, bbl/day

D=Internal diameter, in.

ν=Kinematic viscosity, cSt

Equation (3.16) is simply a modified form of Equation (3.15) after performing conversions to commonly used pipeline units. R is still a dimensionless value.

Another version of the Reynolds number in English units is as follows:

$$R=3160\ Q/(\nu D) \tag{3.17}$$

where

Q=Flow rate, gal/min

D=Internal diameter, in.

ν=Kinematic viscosity, cSt

An equivalent equation for Reynolds number in SI units is

$$R=353{,}678\ Q/(\nu D) \tag{3.18}$$

where

Q=Flow rate, m^3/h

D=Internal diameter, mm

ν=Kinematic viscosity, cSt

As indicated earlier, if the Reynolds number is less than 2000, the flow is considered laminar. This means that the various layers of liquid flow without turbulence in the form of laminations. We now illustrate the various flow regimes using an example.

Consider a 16 in. pipeline, with a wall thickness of 0.250 in., transporting a liquid of viscosity 250 cSt. At a flow rate of 50,000 bbl/day the Reynolds number is, using Equation (3.16),

$$R=92.24(50{,}000)/(250\times15.5)=1190$$

Since R is less than 2000, this flow is laminar. If the flow rate is tripled to 150,000 bbl/day, the Reynolds number becomes 3570 and the flow will be in the critical region. At flow rates above 168,040 bbl/day the Reynolds number exceeds 4000 and the flow will be in the turbulent region. Thus, for this 16 in. pipeline and given liquid viscosity of 250 cSt, flow will be fully turbulent at flow rates above 168,040 bbl/day.

As the flow rate and velocity increase, the flow regime changes. With changes in flow regime, the energy lost due to pipe friction increases. At laminar flow, there is less frictional energy lost compared with turbulent flow.

3.4 Flow Regimes

In summary, the three flow regimes may be distinguished as follows:

Laminar: Reynolds number<2000
Critical: Reynolds number>2000 and Reynolds number<4000
Turbulent: Reynolds number>4000

As liquid flows through a pipeline, energy is lost due to friction between the pipe surface and the liquid and due to the interaction between liquid molecules. This energy lost is at the expense of liquid pressure. (See Equation (2.37), Bernoulli's equation, in Chapter 2.) Hence we refer to the frictional energy lost as the pressure drop due to friction.

The pressure drop due to friction in a pipeline depends on the flow rate, pipe diameter, pipe roughness, liquid specific gravity, and viscosity. In addition, the frictional pressure drop depends on the Reynolds number (and hence the flow regime). Our objective would be to calculate the pressure drop given these pipe and liquid properties and the flow regime.

The pressure drop due to friction in a given length of pipe, expressed in feet of liquid head (h), can be calculated using the Darcy-Weisbach equation as follows:

$$h=f(L/D)(V^2/2g) \tag{3.19}$$

where

f=Darcy friction factor, dimensionless, usually a number between 0.008 and 0.10
L=Pipe length, ft
D=Pipe internal diameter, ft

V=Average liquid velocity, ft/s
g=Acceleration due to gravity, 32.2 ft/s^2 in English units

In laminar flow, the friction factor f depends only on the Reynolds number. In turbulent flow f depends on pipe diameter, internal pipe roughness, and Reynolds number, as we will see shortly.

Example Problem 3.2

Consider a pipeline transporting 4000 bbl/hr of gasoline (Spgr=0.736). Calculate the pressure drop in a 5000 ft length of 16 in. pipe (wall thickness 0.250 in.) using the Darcy-Weisbach equation. Assume the friction factor is 0.02.

Solution

Using Equation (3.10):

$$\text{Average liquid velocity}=0.0119(4000\times24)/(15.5)^2=4.76\ \text{ft/s}$$

Using the Darcy-Weisbach equation (3.19):

$$\text{Pressure drop}=0.02(5000)(12/15.5)(4.76^2/64.4)=27.24\ \text{ft of head}$$

Converting to pressure in psi, using Equation (3.7):

$$\text{Pressure drop}=27.24(0.736)/2.31=8.68\ \text{psi}$$

In the above calculations, the friction factor f was assumed to be 0.02. However, the actual friction factor for a particular flow depends on various factors as explained previously. In the next section, we will see how the friction factor is calculated for the various flow regimes.

3.5 Friction Factor

For laminar flow, with Reynolds number R<2000, the Darcy friction factor f is calculated from the simple relationship

$$f=64/R \tag{3.20}$$

It can be seen from Equation (3.20) that for laminar flow the friction factor depends only on the Reynolds number and is independent of the internal condition of the pipe. Thus, regardless of whether the pipe is smooth or rough, the friction factor for laminar flow is a number that varies inversely with the Reynolds number. Therefore, if the Reynolds number R=1800, the friction factor becomes

$$f=64/1800=0.0356$$

It might appear that, since f for laminar flow decreases with Reynolds number, then from the Darcy-Weisbach equation the pressure drop will decrease with an increase in flow rate. This is not true. Since pressure drop is proportional to the square of the velocity V (Equation 3.19), the influence of V is greater than that of f. Therefore, pressure drop will increase with flow rate in the laminar region.

To illustrate, consider the Reynolds number example in Section 3.3 above. If the flow rate is increased from 50,000 bbl/day to 80,000 bbl/day, the Reynolds number R will increase from 1190 to 1904 (still laminar). The velocity will increase from V_1 to V_2 as follows:

$$V_1 = 0.0119(50{,}000)/(15.5)^2 = 2.48 \text{ ft/s}$$

$$V_2 = 0.0119(80{,}000)/(15.5)^2 = 3.96 \text{ ft/s}$$

Friction factors at 50,000 bbl/day and 80,000 bbl/day flow rate are

$$f_1 = 64/1190 = 0.0538$$

$$f_2 = 64/1904 = 0.0336$$

Considering a 5000 ft length of pipe, the head loss due to friction using the Darcy-Weisbach equation (3.19) is:

$$h_{L1} = 0.0538 \times (5000 \times 12/15.5) \times (2.48^2/64.4) = 19.89 \text{ ft}$$

$$h_{L2} = 0.0336 \times (5000 \times 12/15.5) \times (3.96^2/64.4) = 31.67 \text{ ft}$$

Therefore from the above it is clear in laminar flow that even though the friction factor decreases with a flow increase, the pressure drop still increases with an increase in flow rate.

For turbulent flow, when the Reynolds number R>4000, the friction factor f depends not only on R but also on the internal roughness of the pipe. As the pipe roughness increases, so does the friction factor. Therefore, smooth pipes have a smaller friction factor compared with rough pipes. More correctly, friction factor depends on the relative roughness (e/D) rather than the absolute pipe roughness e.

Various correlations exist for calculating the friction factor f. These are based on experiments conducted by scientists and engineers over the last 60 years or more. A good all-purpose equation for the friction factor f in the turbulent region (i.e., where R>4000) is the Colebrook-White equation:

$$1/\sqrt{f} = -2\mathrm{Log}_{10}[(e/3.7D) + 2.51/(R\sqrt{f})] \tag{3.21}$$

where
  f=Darcy friction factor, dimensionless
  D=Pipe internal diameter, in.
  e=Absolute pipe roughness, in.
  R=Reynolds number of flow, dimensionless

In SI units, the above equation for f remains the same as long as the absolute roughness e and the pipe diameter D are both expressed in mm. All other terms in the equation are dimensionless.

It can be seen from Equation (3.21) that the calculation of f is not easy, since it appears on both sides of the equation. A trial-and-error approach needs to be used. We assume a starting value of f (say, 0.02) and substitute it in the right-hand side of Equation (3.21). This will yield a second approximation for f, which can then be used to re-calculate a better value of f, by successive iteration. Generally, three to four iterations will yield a satisfactory result for f, correct to within 0.001.

During the last two or three decades several formulas for friction factor for turbulent flow have been put forth by various researchers. All these equations attempt to simplify calculation of the friction factor compared with the Colebrook-White equation discussed above. Two such equations that are explicit equations in f, afford easy solution of friction factor compared with the implicit equation (3.21) that requires trial-and-error solution. These are called the Churchill equation and the Swamee-Jain equation and are listed in Appendix C.

In the critical zone, where the Reynolds number is between 2000 and 4000, there is no generally accepted formula for determining the friction factor. This is because the flow is unstable in this region and therefore the friction factor is indeterminate. Most users calculate the value of f based upon turbulent flow.

To make matters more complicated, the turbulent flow region (R>4000) actually consists of three separate regions:

Turbulent flow in smooth pipes
Turbulent flow in fully rough pipes
Transition flow between smooth and rough pipes

For turbulent flow in smooth pipes, pipe roughness has a negligible effect on the friction factor. Therefore, the friction factor in this region depends only on the Reynolds number as follows:

$$1/\sqrt{f} = -2\mathrm{Log}_{10}[2.51/(R\sqrt{f})] \tag{3.22}$$

For turbulent flow in fully rough pipes, the friction factor f appears to be less dependent on the Reynolds number as the latter increases in magnitude. It depends only on the pipe roughness and diameter. It can be calculated from the following equation:

$$1/\sqrt{f} = -2\mathrm{Log}_{10}[(e/3.7D)] \tag{3.23}$$

For the transition region between turbulent flow in smooth pipes and turbulent flow in fully rough pipes, the friction factor f is calculated using the Colebrook-White equation given previously:

$$1/\sqrt{f} = -2\mathrm{Log}_{10}[(e/3.7D) + 2.51/(R\sqrt{f})] \tag{3.24}$$

As mentioned before, in SI units the above equation for f remains the same, provided e and D are both in mm.

The friction factor equations discussed above can also be plotted on a Moody diagram as shown in Figure 3.4. Relative roughness is defined as e/D, and is simply the result of dividing the absolute pipe roughness by the pipe internal diameter. The relative roughness term is a dimensionless parameter. The Moody diagram represents the complete friction factor map for laminar and all turbulent regions of pipe flows. It is used commonly in estimating the friction factor in pipe flow. If the Moody diagram is not available, we must use trial-and-error solution of Equation (3.24) to calculate the friction factor. To use the Moody diagram for

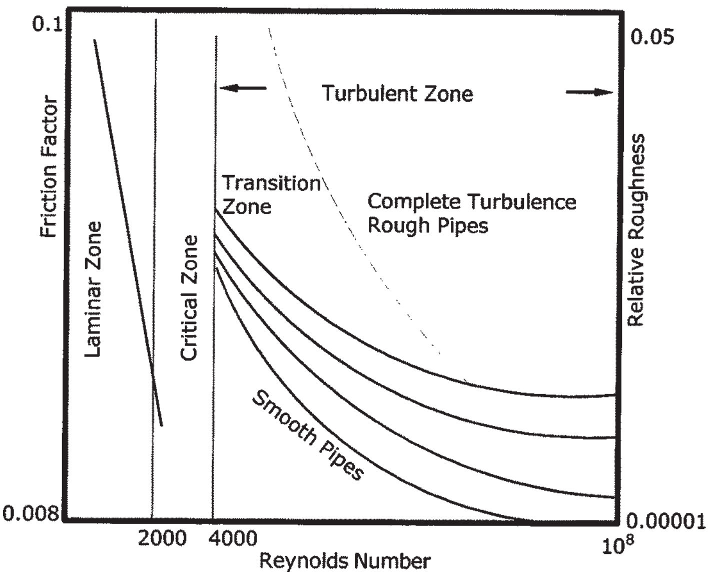

Figure 3.4 Moody diagram for friction factor.

determining the friction factor f we first calculate the Reynolds number R for the flow. Next, we find the location on the horizontal axis of Reynolds number for the value of R and draw a vertical line that intersects with the appropriate relative roughness (e/D) curve. From this point of intersection on the (e/D) curve, we read the value of the friction factor f on the vertical axis on the left.

Before leaving the discussion of the friction factor, we must mention an additional term: the Fanning friction factor. Some publications use this friction factor instead of the Darcy friction factor. The Fanning friction factor is defined as follows:

$$f_f = f_d/4 \tag{3.25}$$

where

f_f=Fanning friction factor

f_d=Darcy friction factor

Unless otherwise specified, we will use the Darcy friction factor throughout this book.

Example Problem 3.3

Water flows through a 20 in. pipe at 5700 gal/min. Calculate the friction factor using the Colebrook-White equation. Assume 0.375 in. pipe wall thickness and an absolute roughness of 0.002 in. Use a specific gravity of 1.00 and a viscosity of 1.0 cSt. What is the head loss due to friction in 2500 ft of pipe?

Solution

First we calculate the Reynolds number from Equation (3.17) as follows:

$$R = 3160 \times 5700/(19.25 \times 1.0) = 935{,}688$$

The flow is fully turbulent and the friction factor f is calculated using Equation (3.21) as follows:

$$1/\sqrt{f} = -2\mathrm{Log}_{10}[(0.002/(3.7 \times 19.25)) + 2.51/(935{,}688\sqrt{f})]$$

The above implicit equation for f must be solved by trial and error. First assume a trial value of f=0.02. Substituting in the equation above, we get successive approximations for f as follows:

f=0.0133, 0.0136, and 0.0136

Therefore the solution is f=0.0136

Using Equation (3.12),

$$\text{Velocity}=0.4085(5700)/19.25^2=6.28\ \text{ft/s}$$

Using Equation (3.19), head loss due to friction is

$$h=0.0136\times(2500\times12/19.25)\times6.28^2/64.4=12.98\ \text{ft}$$

3.6 Pressure Drop due to Friction

In the previous section, we introduced the Darcy-Weisbach equation:

$$h=f(L/D)(V^2/2g) \tag{3.26}$$

where the pressure drop h is expressed in feet of liquid head and the other symbols are defined below

f=Darcy friction factor, dimensionless
L=Pipe length, ft
D=Pipe internal diameter, ft
V=Average liquid velocity, ft/s
g=Acceleration due to gravity, 32.2 ft/s^2 in English units

A more practical equation, using customary pipeline units, is given below for calculating the pressure drop in pipelines. Pressure drop due to friction per unit length of pipe, in English units, is

$$P_m=0.0605fQ^2(Sg/D^5) \tag{3.27}$$

and in terms of transmission factor F

$$P_m=0.2421(Q/F)^2(Sg/D^5) \tag{3.28}$$

where

P_m=Pressure drop due to friction, lb/in.^{2} per mile (psi/mile) of pipe length
Q=Liquid flow rate, bbl/day
f=Darcy friction factor, dimensionless
F=Transmission factor, dimensionless
Sg=Liquid specific gravity
D=Pipe internal diameter, in.

The transmission factor F is directly proportional to the volume that can be transmitted through the pipeline and therefore has an inverse relationship with the friction factor f. The transmission factor F is calculated from the following equation:

$$F=2/\sqrt{f} \tag{3.29}$$

Since the friction factor f ranges from 0.008 to 0.10 it can be seen from Equation (3.29) that the transmission factor F ranges from 6 to 22 approximately.

The Colebrook-White equation (3.21) can be rewritten in terms of the transmission factor F as follows:

$$F=-4\ \mathrm{Log}_{10}[(e/3.7D)+1.255(F/R)]$$

for turbulent flow R>4000 (3.30)

Similar to the calculation of the friction factor f using Equation (3.21), the calculation of transmission factor F from Equation (3.30) will also be a trial-and-error approach. We assume a starting value of F (say 10.0) and substitute it in the right-hand side of Equation (3.30). This will yield a second approximation for F, which can then be used to recalculate a better value, by successive iteration. Generally, three or four iterations will yield a satisfactory result for F.

In SI units, the Darcy equation (in pipeline units) for the pressure drop in terms of the friction factor is represented as follows:

$$P_{km}=6.2475\times10^{10} fQ^2(Sg/D^5) \tag{3.31}$$

and the corresponding equation in terms of transmission factor F is written as follows:

$$P_{km}=24.99\times10^{10}(Q/F)^2(Sg/D^5) \tag{3.32}$$

where

P_{km}=Pressure drop due to friction, kPa/km
Q=Liquid flow rate, m^3/hr
f=Darcy friction factor, dimensionless
F=Transmission factor, dimensionless
Sg=Liquid specific gravity
D=Pipe internal diameter, mm

In SI units, the transmission factor F is calculated using Equation (3.30) as follows:

$$F=-4\mathrm{Log}_{10}[(e/3.7D)+1.255(F/R)]$$

for turbulent flow R>4000 (3.33)

where

D=Pipe internal diameter, mm
e=Absolute pipe roughness, mm
R=Reynolds number of flow, dimensionless

Example Problem 3.4

Consider a 100 mile pipeline, 16 in. diameter, 0.250 in. wall thickness, transporting a liquid (specific gravity of 0.815 and viscosity of 15 cSt at 70°F) at a flow rate of 90,000 bbl/day. Calculate the friction factor and pressure drop per unit length of pipeline using the Colebrook-White equation. Assume a pipe roughness value of 0.002.

Solution

The Reynolds number is calculated first:

$$R = \frac{92.24 \times 90{,}000}{15.5 \times 15} = 35{,}706$$

Using the Colebrook-White equation (3.30), the transmission factor is

$$F=-4\ \mathrm{Log}_{10}[(0.002/(3.7\times 15.5))+1.255F/35{,}706]$$

Solving above equation, by trial and error, yields

$$F=13.21$$

To calculate the friction factor f, we use Equation (3.29) after some transposition and simplification as follows:

$$\text{Friction factor } f=4/F^2=4/(13.21)^2=0.0229$$

The pressure drop per mile is calculated using Equation (3.28):

$$P_m=0.2421(90{,}000/13.21)^2(0.815/15.5^5)=10.24\ \text{psi/mile}$$

The total pressure drop in 100 miles length is then

$$\text{Total pressure drop}=100\times 10.24=1024\ \text{psi}$$

3.7 Colebrook-White Equation

In 1956 the U.S. Bureau of Mines conducted experiments and recommended a modified version of the Colebrook-White equation. The modified Colebrook-White equation yields a more conservative transmission factor F. The pressure drop calculated using the modified Colebrook-White equation is slightly higher than that calculated using the original Colebrook-White equation. This modified Colebrook-White equation, in terms of transmission factor F, is defined as follows:

$$F=-4\mathrm{Log}_{10}[(e/3.7D)+1.4125(F/R)] \tag{3.34}$$

In SI units, the transmission factor equation above remains the same with e and D expressed in mm, other terms being dimensionless.

Comparing Equation (3.34) with Equation (3.30) or (3.33), it can be seen that the only change is in the substitution of the constant 1.4125 in place of 1.255 in the original Colebrook-White equation. Some companies use the modified Colebrook-White Equation stated in Equation (3.34).

An explicit form of an equation to calculate the friction factor was proposed by Swamee and Jain. This equation does not require trial-and-error solution like the Colebrook-White equation. It correlates very closely with the Moody diagram values. Appendix C gives a version of the Swamee-Jain equation for friction factor.

3.8 Hazen-Williams Equation

The Hazen-Williams equation is commonly used in the design of water distribution lines and in the calculation of frictional pressure drop in refined petroleum products such as gasoline and diesel. This method involves the use of the Hazen-Williams C-factor instead of pipe roughness or liquid viscosity. The pressure drop calculation using the Hazen-Williams equation takes into account flow rate, pipe diameter, and specific gravity as follows:

$$h=4.73L(Q/C)^{1.852}/D^{4.87} \tag{3.35}$$

where

h=Head loss due to friction, ft
L=Pipe length, ft
D=Pipe internal diameter, ft
Q=Flow rate, ft^3/s
C=Hazen-Williams coefficient or C-factor, dimensionless

Typical values of the Hazen-Williams C-factor are given in Appendix A, Table A.8.

In customary pipeline units, the Hazen-Williams equation can be rewritten as follows in English units:

$$Q=0.1482(C)(D)^{2.63}\ (P_m/Sg)^{0.54} \tag{3.36}$$

where

Q=Flow rate, bbl/day
D=Pipe internal diameter, in.
P_m=Frictional pressure drop, psi/mile
Sg=Liquid specific gravity
C=Hazen-Williams C-factor

Another form of Hazen-Williams equation, when the flow rate is in gal/min and head loss is measured in feet of liquid per thousand feet of pipe, is as follows:

$$GPM=6.7547\times10^{-3}(C)(D)^{2.63}(H_L)^{0.54} \tag{3.37}$$

where

GPM=Flow rate, gal/min

H_L=Friction loss, ft of liquid per 1000 ft of pipe

Other symbols are as in Equation (3.36).

In SI units, the Hazen-Williams equation is as follows:

$$Q=9.0379\times10^{-8}(C)(D)^{2.63}(P_{km}/Sg)^{0.54} \tag{3.38}$$

where

Q=Flow rate, m^3/hr

D=Pipe internal diameter, mm

P_{km}=Frictional pressure drop, kPa/km

Sg=Liquid specific gravity

C=Hazen-Williams C-factor

Historically, many empirical formulas have been used to calculate frictional pressure drop in pipelines. The Hazen-Williams equation has been widely used in the analysis of pipeline networks and water distribution systems because of its simple form and ease of use. A review of the Hazen-Williams equation shows that the pressure drop due to friction depends on the liquid specific gravity, pipe diameter, and the Hazen-Williams coefficient or C-factor.

Unlike the Colebrook-White equation, where the friction factor is calculated based on pipe roughness, pipe diameter, and the Reynolds number, which further depends on liquid specific gravity and viscosity, the Hazen-Williams C-factor appears not to take into account the liquid viscosity or pipe roughness. It could be argued that the C-factor is in fact a measure of the pipe internal roughness. However, there does not seem to be any indication of how the C-factor varies from laminar flow to turbulent flow.

We could compare the Darcy-Weisbach equation with the Hazen-Williams equation and infer that the C-factor is a function of the Darcy friction factor and Reynolds number. Based on this comparison it can be concluded that the C-factor is indeed an index of relative roughness of the pipe. It must be remembered that the Hazen-Williams equation, though convenient from the standpoint of its explicit nature, is an empirical equation, and also that it is difficult to apply to all fluids under all conditions. Nevertheless, in real-world pipelines, with sufficient field data we could determine specific C-factors for specific pipelines and fluids pumped.

Example Problem 3.5

A 3 in. (internal diameter) smooth pipeline is used to pump 100 gal/min of water. Using the Hazen-Williams equation, calculate the head loss in 3000 ft of this pipe. Assume a C-factor of 140.

Solution

Using Equation (3.37), substituting given values, we get

$$100=6.7547\times10^{-3}\times140(3.0)^{2.63}(H_L)^{0.54}$$

Solving for the head loss we get

$$H_L=26.6 \text{ ft per } 1000 \text{ ft}$$

Therefore head loss for 3000 ft=26.6×3=79.8 ft of water.

3.9 Shell-MIT Equation

The Shell-MIT equation, sometimes called the MIT equation, is used in the calculation of pressure drop in heavy crude oil and heated liquid pipelines. Using this method, a modified Reynolds number Rm is calculated first from the Reynolds number as follows:

$$R=92.24(Q)/(D\nu) \tag{3.39}$$

$$Rm=R/(7742) \tag{3.40}$$

where
R=Reynolds number, dimensionless
Rm=Modified Reynolds number, dimensionless
Q=Flow rate, bbl/day
D=Pipe internal diameter, in.
ν=Kinematic viscosity, cSt

Next, depending on the flow (laminar or turbulent), the friction factor is calculated from one of the following equations:

$$f=0.00207/Rm \text{ (laminar flow)} \tag{3.41}$$

$$f=0.0018+0.00662(1/Rm)^{0.355} \text{ (turbulent flow)} \tag{3.42}$$

Note that this friction factor f in Equations (3.41) and (3.42) is not the same as the Darcy friction factor f discussed earlier. In fact, the friction factor f in the above equations is more like the Fanning friction factor discussed previously.

Finally, the pressure drop due to friction is calculated using the equation

$$P_m = 0.241(f\ SgQ^2)/D^5 \tag{3.43}$$

where
P_m=Frictional pressure drop, psi/mile
f=Friction factor, dimensionless
Sg=Liquid specific gravity
Q=Flow rate, bbl/day
D=Pipe internal diameter, in.
In SI units the MIT equation is expressed as follows:

$$P_m = 6.2191\times10^{10}(f\ SgQ^2)/D^5 \tag{3.44}$$

where
P_m=Frictional pressure drop, kPa/km
f=Friction factor, dimensionless
Sg=Liquid specific gravity
Q=Flow rate, m^3/hr
D=Pipe internal diameter, mm
Comparing Equation (3.43) with Equations (3.27) and (3.28) and recognizing the relationship between transmission factor F and Darcy friction factor f, using Equation (3.29) it is evident that the friction factor f in Equation (3.43) is not the same as the Darcy friction factor. It appears to be one-fourth the Darcy friction factor.

Example Problem 3.6

A steel pipeline of 500 mm outside diameter, 10 mm wall thickness is used to transport heavy crude oil at a flow rate of 800 m3/hr at 100°C. Using the MIT equation calculate the friction loss per kilometer of pipe assuming an internal pipe roughness of 0.05 mm. The heavy crude oil has a specific gravity of 0.89 at 100°C and a viscosity of 120 cSt at 100°C.

Solution

From Equation (3.18)

$$\text{Reynolds number} = 353{,}678\times800/(120\times480) = 4912$$

The flow is therefore turbulent.

$$\text{Modified Reynolds number} = 4912/7742 = 0.6345$$

$$\text{Friction factor} = 0.0018 + 0.00662(1/0.6345)^{0.355} = 0.0074$$

Pressure drop from Equation (3.44) is

$$P_m=6.2191\times10^{10}(0.0074\times0.89\times800\times800)/480^5$$
$$=10.29 \text{ kPa/km}$$

3.10 Miller Equation

The Miller equation, also known as the Benjamin Miller formula, is used in hydraulics studies involving crude oil pipelines. This equation does not consider pipe roughness and is an empirical formula for calculating the flow rate from a given pressure drop. The equation can also be rearranged to calculate the pressure drop from a given flow rate. One of the popular versions of this equation is as follows:

$$Q=4.06(M)(D^5P_m/Sg)^{0.5} \tag{3.45}$$

where M is defined as follows:

$$M=\text{Log}_{10}(D^3SgP_m/cp^2)+4.35 \tag{3.46}$$

and

Q=Flow rate, bbl/day
D=Pipe internal diameter, in.
P_m=Frictional pressure drop, psi/mile
Sg=Liquid specific gravity
cp=Liquid viscosity, centipoise

In SI units, the Miller equation is as follows:

$$Q=3.996\times10^{-6}(M)(D^5P_m/Sg)^{0.5} \tag{3.47}$$

where M is defined as follows:

$$M=\text{Log}_{10}(D^3SgP_m/cp^2)-0.4965 \tag{3.48}$$

where

Q=Flow rate, m^3/hr
D=Pipe internal diameter, mm
P_m=Frictional pressure drop, kPa/km
Sg=Liquid specific gravity
cp=Liquid viscosity, centipoise

It can be seen from the above version of the Miller equation that calculating the pressure drop P_m from the flow rate Q is not straightforward. This is because the parameter M depends on the pressure drop P_m. Therefore, if we solve for P_m in terms of Q and other parameters from Equation (3.45), we get

$$P_m = (Q/4.06M)^2 (Sg/D^5) \quad (3.49)$$

where M is calculated from Equation (3.46).

To calculate P_m from a given value of flow rate Q, we use a trial-and-error approach. First, we assume a value of P_m to get a starting value of M from Equation (3.46). This value of M is then substituted in Equation (3.49) to determine a second approximation for P_m. This value of P_m will be used to generate a better value of M from Equation (3.46) which is then used to recalculate P_m. Once the successive values of P_m are within an allowable tolerance, such as 0.01 psi/mile, the iteration can be terminated and the value of pressure drop P_m has been calculated.

Example Problem 3.7

Using the Miller equation determine the pressure drop in a 14 in. diameter, 0.250 in. wall thickness, crude oil pipeline at a flow rate of 3000 gal/min. The crude oil specific gravity is 0.825 at 60°F and the viscosity is 15 cSt at 60°F.

Solution

Liquid viscosity in centipoise=0.825×15=12.375 cP

First the parameter M is calculated from Equation (3.46) using an initial value of P_m=10.0:

$$M = \mathrm{Log}_{10}(13.5^3 \times 0.825 \times 10.0/12.375^2) + 4.35$$

$$= 6.4724$$

Using this value of M in Equation (3.49), we get

$$P_m = [3000 \times 34.2857/(4.06 \times 6.4724)]^2 (0.825/13.5^5)$$

$$= 28.19 \text{ psi/mile}$$

We were quite far off in our initial estimate of P_m.
Using this value of P_m, a new value of M is calculated as

$$M = 6.9225$$

Substituting this value of M in Equation (3.49), we get

$$P_m = 24.64$$

By successive iteration, we get a final value for P_m of 25.02 psi/mile.

3.11 T.R.Aude Equation

Another pressure drop equation used in the pipeline industry that is popular among companies that transport refined petroleum products is the T.R.Aude equation, sometimes referred to simply as the Aude equation. This equation is named after the engineer who conducted experiments on pipelines in the 1950s.

The Aude equation is used in pressure drop calculations for 8 in. to 12 in. pipelines. This method requires the use of the Aude K-factor, representing pipeline efficiency. One version of this formula is given below:

$$P_m=[Q(z^{0.104})(Sg^{0.448})/(0.871(K)(D^{2.656}))]^{1.812} \tag{3.50}$$

where

P_m=Pressure drop due to friction, psi/mile
Q=Flow rate, bbl/hr
D=Pipe internal diameter, in.
Sg=Liquid specific gravity
z=Liquid viscosity, centipoise
K=T.R.Aude K-factor, usually 0.90 to 0.95

In SI units the Aude equation is as follows:

$$P_m=8.888\times10^8[Q(z^{0.104})(Sg^{0.448})/(K(D^{2.656}))]^{1.812} \tag{3.51}$$

where

P_m=Frictional pressure drop, kPa/km
Sg=Liquid specific gravity
Q=Flow rate, m^3/hr
D=Pipe internal diameter, mm
z=Liquid viscosity, centipoise
K=T.R.Aude K-factor, usually 0.90 to 0.95

Since the Aude equation for pressure drop given above does not contain pipe roughness, it can be deduced that the K-factor somehow must take into account the internal condition of the pipe. As with the Hazen-Williams C-factor discussed earlier, the Aude K-factor is also an experience-based factor and must be determined by field measurement and calibration of an existing pipeline. If field data is not available, engineers usually approximate using a value such as K=0.90 to 0.95. A higher value of K will a result in a lower pressure drop for a given flow rate or a higher a flow rate for a given pressure drop.

It must be noted that the Aude equation is based on field data collected from 6 in. and 8 in. refined products pipelines. Therefore, caution must be used when applying this formula to larger pipelines.

3.12 Minor Losses

In most long-distance pipelines, such as trunk lines, the pressure drop due to friction in the straight lengths of pipe forms the significant proportion of the total frictional pressure drop. Valves and fittings contribute very little to the total pressure drop in the entire pipeline. Hence, in such cases, pressure losses through valves, fittings, and other restrictions are generally classified as *"minor losses"*. Minor losses include energy losses resulting from rapid changes in the direction or magnitude of liquid velocity in the pipeline. Thus pipe enlargements, contractions, bends, and restrictions such as check valves and gate valves are included in minor losses.

In short pipelines, such as terminal and plant piping, the pressure loss due to valves, fittings, etc., may be a substantial portion of the total pressure drop. In such cases, the term "minor losses" is a misnomer.

Therefore, in long pipelines the pressure losses through bends, elbows, valves, fittings, etc., are classified as "minor" and in most instances may be neglected without significant error. However, in shorter pipelines these losses must be included for correct engineering calculations. Experiments with water at high Reynolds numbers have shown that the minor losses vary approximately as the square of the velocity. This leads to the conclusion that minor losses can be represented by a function of the liquid velocity head or kinetic energy ($V^2/2g$).

Accordingly, the pressure drop through valves and fittings is generally expressed in terms of the liquid kinetic energy $V^2/2g$ multiplied by a head loss coefficient K. Comparing this with the Darcy-Weisbach equation for head loss in a pipe, we can see the following analogy. For a straight pipe, the head loss h is $V^2/2g$ multiplied by the factor (fL/D). Thus, the head loss coefficient for a straight pipe is fL/D.

Therefore, the pressure drop in a valve or fitting is calculated as follows:

$$h=KV^2/2g \tag{3.52}$$

where

h=Head loss due to valve or fitting, ft

K=Head loss coefficient for the valve or fitting, dimensionless

V=Velocity of liquid through valve or fitting, ft/s

g=Acceleration due to gravity, 32.2 ft/s^2 in English units

The head loss coefficient K is, for a given flow geometry, considered practically constant at high Reynolds number. K increases with pipe roughness and with lower Reynolds numbers. In general the value of K is determined mainly by the flow geometry or by the shape of the pressure loss device.

It can be seen from Equation (3.52) that K is analogous to the term (fL/D) for a straight length of pipe. Values of K are available for various types of valves and fittings in standard handbooks, such as the Crane Handbook *Flow of Fluids through Valves, Fittings, and Pipes* and *Cameron Hydraulic Data*. A table of K-factor values commonly used for valves and fittings is included in Appendix A, Table A.9.

3.12.1 Gradual Enlargement

Consider liquid flowing through a pipe of diameter D_1. If at a certain point the diameter enlarges to D_2, the energy loss that occurs due to the enlargement can be calculated as follows:

$$h=K(V_1-V_2)^2/2g \tag{3.53}$$

where V_1 and V_2 are the velocity of the liquid in the smaller-diameter and the larger-diameter pipe respectively. The value of K depends upon the diameter ratio D_1/D_2 and the different cone angle due to the enlargement. A gradual enlargement is shown in Figure 3.5.

For a sudden enlargement K=1.0 and the corresponding head loss is

$$h=(V_1-V_2)^2/2g \tag{3.54}$$

Example Problem 3.8

Calculate the head loss due to a gradual enlargement in a pipe that flows 100 gal/min of water from a 2 in. diameter to a 3 in. diameter with an included angle of 30°. Both pipe sizes are internal diameters.

Solution

The liquid velocities in the two pipe sizes are as follows:

$$V_1=0.4085\times100/2^2=10.21 \text{ ft/s}$$

$$V_2=0.4085\times100/3^2=4.54 \text{ ft/s}$$

Diameter ratio=3/2=1.5

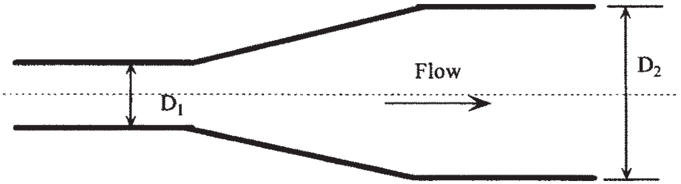

Figure 3.5 Gradual enlargement.

From charts, for diameter ratio=1.5 and cone angle=30° the value of K is

$K=0.38$

Therefore head loss due to gradual enlargement is

$h=0.38\times(10.21-4.54)^2/64.4=0.19$ ft

If the expansion were a sudden enlargement from 2 in. to 3 in., the head loss would be

$h=(10.21-4.54)^2/64.4=0.50$ ft

3.12.2 Abrupt Contraction

For flow through an abrupt contraction, the flow from the larger pipe (diameter D_1 and velocity V_1) to a smaller pipe (diameter D_2 and velocity V_2) results in the formation of a vena contracta or throat, immediately after the diameter change. At the vena contracta, the flow area reduces to A_c with increased velocity of V_c. Subsequently the flow velocity decreases to V_2 in the smaller pipe. Thus from velocity V_1, the liquid first accelerates to velocity V_c at the vena contracta and subsequently decelerates to V_2. This is shown in Figure 3.6.

The energy loss due to the sudden contraction depends upon the ratio of the pipe diameters D_2 and D_1 and the ratio A_c/A_2. The value of the head loss coefficient K can be found using Table 3.1, where $C_c=A_c/A_2$. The ratio A_2/A_1 can be calculated from the ratio of the diameters D_2/D_1.

A pipe connected to a large storage tank represents a type of abrupt contraction. If the storage tank is a large body of liquid, we can state that this is a limiting case of the abrupt contraction. For such a square-edged pipe entrance from a large tank $A_2/A_1=0$. From Table 3.1, for this case K=0.5 for turbulent flow.

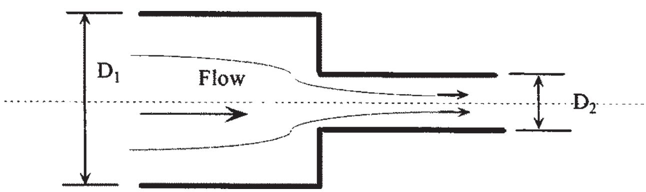

Figure 3.6 Abrupt contraction.

Table 3.1 Head Loss Coefficient K for Abrupt Contraction

A_2/A_1	C_c	K
0.0	0.617	0.50
0.1	0.624	0.46
0.2	0.632	0.41
0.3	0.643	0.36
0.4	0.659	0.30
0.5	0.681	0.24
0.6	0.712	0.18
0.7	0.755	0.12
0.8	0.813	0.06
0.9	0.892	0.02
1.0	1.000	0.00

Another type of pipe entrance from a large tank is called a re-entrant pipe entrance. If the pipe is thin-walled and the opening within the tank is located more than one pipe diameter upstream from the tank wall, the K value will be close to 0.8.

If the edges of the pipe entrance in a tank are rounded or bell-shaped, the head loss coefficient is considerably smaller. An approximate value for K for a bell-mouth entrance is 0.1.

3.12.3 Head Loss and L/D Ratio for Pipes and Fittings

We have discussed how minor losses can be accounted for using the head loss coefficient K in conjunction with the liquid velocity head. Table A.9 in Appendix A lists K values for common valves and fittings.

Referring to the Table A.9 for K values, we see that for a 16 in. gate valve, K=0.10. Therefore, compared with a 16 in. straight pipe, we can write, from the Darcy-Weisbach equation:

$$\frac{fL}{D} = 0.10$$

or

$$\frac{L}{D} = 0.10f$$

If we assume f=0.0125, we get

$$\frac{L}{D} = 8$$

This means that compared with a straight pipe 16 in. in diameter, a 16 in. gate valve has an L/D ratio of 8, which causes the same friction loss. The L/D ratio represents the equivalent length of straight pipe in terms of its diameter that will equal the pressure loss in the valve or fitting. In Table A.10 in Appendix A the L/D ratios for various valves and fittings are given.

Using the L/D ratio we can replace a 16 in. gate valve with 8×16 in.= 128 in. of straight 16 in. pipe. This length of pipe will have the same friction loss as the 16 in. gate valve. Thus we can use the K values or L/D ratios to calculate the friction loss in valves and fittings.

3.13 Internally Coated Pipes and Drag Reduction

In turbulent flow, pressure drop due to friction depends on the pipe roughness. Therefore, if the internal pipe surface can be made smoother, the frictional pressure drop can be reduced. Internally coating a pipeline with an epoxy coating will considerably reduce the pipe roughness compared with uncoated pipe.

For example, if the uncoated pipe has an absolute roughness of 0.002 in., coating the pipe can reduce roughness to a value as low as 0.0002 in. The friction factor f may therefore reduce from 0.02 to 0.01 depending on the flow rate, Reynolds number, etc. Since pressure drop is directly proportional to the friction factor in accordance with the Darcy-Weisbach equation, the total pressure drop in the internally coated pipeline in this example would be 50% of that in the uncoated pipeline.

Another method of reducing frictional pressure drop in a pipeline is by using drag reduction. Drag reduction is the process of reducing the pressure drop due to friction in a pipeline by continuously injecting a very small quantity (parts per million, or ppm) of a high-molecular-weight hydrocarbon, called the drag reduction agent (DRA), into the flowing liquid stream. The DRA is effective only in pipe segments between two pump stations. It degrades in performance as it flows through the pipeline for long distances. It also completely breaks up or suffers shear degradation as it passes through pump stations, meters and other restrictions. DRA works only in turbulent flow and with low-viscosity liquids. Thus, it works well with refined petroleum products (gasoline, diesel, etc.) and light crude oils. It is ineffective in heavy crude oil pipelines, particularly in laminar flow. Currently, the two leading vendors of DRA products in the United States are Baker Petrolite and Conoco-Phillips.

To determine the amount of drag reduction using DRA we proceed as follows. If the pressure drops due to friction with and without DRA are known, we can calculate the percentage drag reduction:

$$\text{Percentage drag reduction}=100(DP_0-DP_1)/DP_0 \qquad (3.55)$$

where

DP_0=Friction drop in pipe segment without DRA, psi

DP_1=Friction drop in pipe segment with DRA, psi

The above pressure drops are also referred to as untreated versus treated pressure drops. It is fairly easy to calculate the value of untreated pressure drop, using the pipe size, liquid properties, etc. The pressure drop with DRA is obtained using DRA vendor information. In most cases involving DRA, we are interested in calculating how much DRA we need to use to reduce the pipeline friction drop, and hence the pumping horsepower required. It must be noted that DRA may not be effective at the higher flow rate, if existing pump and driver limitations preclude operating at higher flow rates due to pump driver horsepower limitation.

Consider a situation where a pipeline is limited in throughput due to maximum allowable operating pressures (MAOP). Let us assume the friction drop in this MAOP limited pipeline is 800 psi at 100,000 bbl/day. We are interested in increasing pipeline flow rate to 120,000 bbl/day using DRA and we would proceed as follows:

Flow improvement desired=(120,000-100,000)/100,000=20%

If we calculate the actual pressure drop in the pipeline at the increased flow rate of 120,000 bbl/day (ignoring the MAOP violation) and assume that we get the following pressure drop:

Frictional pressure drop at 120,000 bbl/day=1150 psi

and

Frictional pressure drop at 100,000 bbl/day=800 psi

The percentage drag reduction is then calculated from Equation (3.55) as

Percentage drag reduction=100(1150-800)/1150=30.43%

In the above calculation we have tried to maintain the same frictional drop (800 psi) using DRA at the higher flow rate as the initial pressure-limited case. Knowing the percentage drag reduction required, we can get the DRA vendor to tell us how much DRA will be required to achieve the 30.43% drag reduction, at the flow rate of 120,000 bbl/day. If the answer is 15 ppm of Brand X DRA, we can calculate the daily DRA requirement as follows:

Quantity of DRA required=$(15/10^6)(120{,}000)(42)$=75.6 gal/day

If DRA costs $10 per gallon, this equates to a daily DRA cost of $756. In this example, a 20% flow improvement requires a drag reduction of 30.43% and 15 ppm of DRA, costing $756 per day. Of course, these are simply rough numbers, used to illustrate the DRA calculations methods. The quantity of DRA required will depend on the pipe size, liquid viscosity, flow rate, and Reynolds number, in addition to the percentage drag reduction required. Most DRA vendors will confirm that drag reduction is effective only in turbulent flow (Reynolds number>4000) and that it does not work with heavy (high-viscosity) crude oil and other liquids.

Also, drag reduction cannot be increased indefinitely by injecting more DRA. There is a theoretical limit to the drag reduction attainable. For a certain range of flow rates, the percentage drag reduction will increase as the DRA ppm is increased. At some point, depending on the pumped liquid, flow characteristics, etc., the drag reduction levels off. No further increase in drag reduction is possible by increasing the DRA ppm. We would have reached the point of diminishing returns, in this case.

In Chapter 12 on feasibility studies and pipeline economics, we explore the subject of DRA further.

3.14 Summary

We have defined pressure and how it is measured in both a static and dynamic context. The velocity and Reynolds number calculations for pipe flow were introduced and the use of the Reynolds number in classifying liquid flow as laminar, critical, and turbulent were explained. Existing methods of calculating the pressure drop due to friction in a pipeline using the Darcy-Weisbach equation were discussed and illustrated using examples. The importance of the Moody diagram was explained. Also, the trial-and-error solutions of friction factor from the Colebrook-White equation were covered. The use of the Hazen-Williams, MIT and other pressure drop equations were discussed. Minor losses in pipelines due to valves, fittings, pipe enlargements, and pipe contractions were analyzed. The concept of drag reduction as a means of reducing frictional head loss was also introduced.

3.15 Problems

3.15.1 Calculate the average velocity and Reynolds number in a 20 in. pipeline that transports diesel fuel at a flow rate of 250,000 bbl/day. Assume 0.375 in. pipe wall thickness and the diesel fuel properties as follows: Specific gravity=0.85, Kinematic viscosity=5.9 cSt. What is the flow regime in this case?

3.15.2 In the above example, what is the value of the Darcy friction factor using the Colebrook-White equation? If the modified Colebrook-White equation is used, what is the difference in friction factors? Calculate the pressure drop due to friction in a 5 mile segment of this pipeline. Use a pipe roughness value of 0.002 in.

3.15.3 Using the Hazen-Williams equation with a C-factor of 125, calculate the frictional pressure drop per mile in the 20 in. pipeline described in Problem 3.15.1 above. Repeat the calculations using the Shell-MIT and Miller equations.

3.15.4 A crude oil pipeline 500 km long with 400 mm outside diameter and 8 mm wall thickness is used to transport 600 m^3/hr of product from a crude oil terminal at San José to a refinery located at La Paz. Assuming the crude oil has a specific gravity of 0.895 and viscosity of 200 SSU at 20°C, calculate the total pressure drop due to friction in the pipeline. If the MAOP of the pipeline is limited to 10 MPa, how many pumping stations will be required to transport this volume, assuming flat terrain? Use the modified Colebrook-White equation and assume a value of 0.05 mm for the absolute pipe roughness.

3.15.5 In Problem 3.15.4 the volume transported was 600 m^3/hr. It is desired to increase flow rate using DRA in the bottleneck section of the pipeline.

(a) What is the maximum throughput possible with DRA?
(b) Summarize any changes needed to the pump stations to handle the increased throughput.
(c) What options are available to further increase pipeline throughput?